

Article

Temporal Event-Causality Graphs for Financial Contagion: Learning Shock Propagation Across Event Types from Financial News

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Abstract

Financial contagion—the propagation of shocks across markets, sectors, and time—remains one of the central questions in empirical finance, and yet most computational approaches model it at the wrong granularity. Existing work captures contagion at the asset level by measuring realised return or volatility comovements between specific securities, which conflates a structural pattern with its surface manifestation. We argue that the genuine causal regularity in contagion is between event types (rate decisions, default announcements, regulatory actions) rather than between individual assets. This paper introduces TECG, a framework that **learns time-stamped causal-temporal association edges—predictive dependencies rather than identified causal effects—between abstract financial event types** from news text and uses them to forecast multi-step contagion cascades. Methodologically, TECG can be read as a neural impulse response function (IRF) for event sequences: where the canonical structural-VAR IRF traces the dynamic response of one continuous variable to a shock in another, TECG traces the conditional intensity with which one event type fires after a shock to another event type, at a learned and possibly multi-modal lag. The framework couples LLM-based event extraction with a multivariate neural Hawkes process whose intensity functions are parameterised by a temporal graph neural network. The resulting graph has interpretable, time-stamped edges of the form “event-type A triggers event-type B with lag distribution L and conditional intensity κ .” We evaluate TECG on an aggregated corpus of approximately 482,000 financial news items spanning 2007–2023 and report three findings. First, **the learned edges recover associations consistent with known causal relationships in finance**—central-bank announcements preceding sector-level earnings revisions, default events at one institution preceding due-diligence events at competitors

—without supervision on these relationships. Second, edges learned from data up to 2018 transfer to held-out cascades from 2020 (COVID equity crash) and 2023 (US regional banking stress) with substantially better fidelity than baselines that ignore temporal structure or operate at the asset level. Third, the framework supports generalised impulse response analysis: given a hypothetical seed event, it produces a distribution over downstream event chains that an analyst can interrogate, time-stamp, and entity-resolve. We are explicit that the empirical validation is observational and that the causal interpretation rests on assumptions we discuss at length. All headline comparisons are supported by paired-bootstrap significance tests; the extraction front end is independently evaluated on a manually annotated benchmark, with an explicit error-propagation analysis; and additional comparisons against recent temporal point-process and temporal-graph baselines are reported. The framework’s principal limitation is that it does not separately identify endogenous fire-sale or leverage-spiral dynamics; we point to where these gaps could be closed.

Keywords: Financial contagion, impulse response function, temporal point processes, graph neural networks, event extraction, causal inference, financial NLP, neural Hawkes process.

1. Introduction

Financial markets have a long history of behaving in ways that look catastrophically correlated only in retrospect. The 1998 collapse of Long-Term Capital Management spread from Russian government debt to United States equity volatility through positions that nobody at the time would have described as connected. The 2008 crisis propagated through securitisation channels that prudential supervisors had not modelled. In early 2020 a respiratory virus in Wuhan moved equity prices in Brazil within days. In March 2023 the failure of one regional bank in California created funding stress for unrelated institutions on three continents inside a single week.

Each of these episodes prompted a recurring question among regulators, academics, and risk managers: given a shock at a particular point in the financial system, can we predict how it will propagate? The literature on systemic risk and financial contagion has produced a substantial answer at the level of asset prices [1]–[10]. Variance-decomposition spillover indices, network DebtRank measures, and asymmetric volatility transmission models have become standard tools. They share a common architecture—

model the time-varying covariance of asset returns, possibly conditioned on a network of exposures, and infer propagation from those covariances.

This paper takes a different starting point. We observe that the genuine causal regularity in contagion is not between specific assets but between event types. A rate-hike announcement does not have an idiosyncratic effect on the Nasdaq Composite; it has a structural effect on rate-sensitive sectors that happens to manifest in the Nasdaq because of its composition. A bank-run event at one institution does not propagate to other institutions because the institutions are linked in some statically defined sense; it propagates because bank-run events at one entity tend to generate due-diligence events at similar entities, which tend to trigger withdrawal events at depositors, which produce funding-stress events at firms with shared deposit demographics. The contagion is over event types, mediated by latent entity attributes, and lagged in characteristic ways that depend on the channels involved.

If this intuition is right—and we present empirical evidence below that it is, with caveats we are careful about—then the appropriate object to learn is a graph whose nodes are abstract financial event types and whose edges encode the conditional intensity with which one event type triggers another, parameterised by the entities and lags involved. We call this object a temporal event-causality graph. Building one is non-trivial. It requires three pieces that do not yet exist as an integrated system: extraction of structured event representations from financial news at scale, learning of causal-temporal edges from observed event sequences subject to all the well-known difficulties of inferring causality from observational data, and a propagation mechanism that can take a newly observed seed event and forecast its downstream cascade with calibrated uncertainty.

A natural objection is that systems with similar ambitions already exist. Quantitative funds use proprietary event-tagged news feeds such as RavenPack and MarketPsych together with regression-based event-study methodology going back several decades. We share the underlying motivation but believe two things differentiate the present effort. First, existing event-study work operates at the asset-return level, mapping events directly to price moves; it does not learn an intermediate causal structure between event types that an analyst could inspect, query, or counterfactually probe. Second, modern temporal point-process and temporal-graph methods have matured to the point where the integration is now feasible, yet the financial-events application has not been built and tested. Recent work has applied temporal graph networks to stock-return forecasting [9], [10] and to event-driven trading signals [20], but to our knowledge no published paper learns event-type-level causal graphs from text-extracted events and validates them on out-of-sample contagion episodes.

A second framing worth naming explicitly is the impulse response function (IRF) tradition in econometrics. Since Sims [5] introduced structural vector autoregressions as a tool for tracing how shocks propagate through macroeconomic time series, the IRF has been the canonical object through which economists ask "what happens to Y at horizon h , if X is shocked by one unit at time 0?" Pesaran and Shin [6] later generalised this to settings where shocks are not orthogonalised, removing the dependence on a contrived ordering of variables. The kernel $\kappa_{\{a \rightarrow b\}}(\tau)$ that TECG learns is, in structural terms, an impulse response function: it traces how the conditional intensity of event-type b responds, as a function of lag τ , to an impulse of event-type a . The framework's contribution is to bring this IRF apparatus to a regime where the standard SVAR-IRF does not directly apply—where the response variable is the timing and type of a future event rather than the value of a continuous return series, where the dimensionality is in the hundreds of event types rather than a handful of macro variables, and where the underlying data is text rather than pre-cleaned numeric series. We return to this connection in Section III, where we formalise the equivalence, and in Section IV, where we discuss what the IRF perspective preserves and where it has to be generalised.

Our central contribution is not in any single component. Hawkes processes have existed for half a century [11]. Temporal graph networks have a five-year history [13]–[15]. Event extraction from financial text is the topic of our previous work and an active research area [15], [16]. The contribution is in showing that these can be coupled productively, and in establishing what is empirically true once they are. Specifically, we make three claims, listed in increasing order of ambition and decreasing order of confidence.

Type-level causal-temporal associations are learnable from extracted financial events. The neural-Hawkes-on-graph machinery we present recovers edges that correspond to known causal patterns in finance—central-bank tightening preceding sector-level revisions, default events at one institution preceding due-diligence events at competitors—without supervision on these patterns. This is a measurement claim, but it is a measurement that nobody has previously made at this granularity, and we believe it stands on its own.

Causal structure transfers across episodes. Edges trained on news up to 2018 retain non-trivial predictive value for cascades observed in 2020 (the COVID equity crash) and 2023 (the United States regional banking stress). This is the empirically substantive claim of the paper. It is also the claim most likely to fail under stricter evaluation, and we report results that distinguish the cases where transfer works from those where it does not.

The framework supports interpretable counterfactual simulation. Given a hypothetical seed event such as a surprise 75-basis-point hike at the next FOMC meeting,

the system produces a distribution over downstream event chains that is time-stamped and entity-resolved. This is the contribution most likely to be useful operationally. It is also the one we are most cautious about empirically, because counterfactual validity in observational settings is genuinely hard to establish and we cannot fully address it without intervention data we do not possess.

We are deliberately modest about scope. We do not claim that our system replaces existing risk-management infrastructure, nor that it captures every form of contagion. Endogenous fire-sale dynamics, leverage spirals, and counterparty-network effects are present in the data we model but are not separately identified by our framework; we treat their decomposition as future work and discuss it in Section VII. We also do not claim a unique novelty for the individual components we use. Section II is therefore organised around what is and is not new in each ingredient, before we describe how they are combined.

The remainder of the paper is organised as follows. Section II reviews four bodies of literature we draw from: financial contagion modelling, temporal point processes, temporal graph networks, and financial NLP, with explicit discussion of what each contributes and where it falls short for our purpose. Section III formalises the temporal event-causality problem and states our assumptions. Section IV describes the proposed framework. Section V details the experimental setup, including the news corpus, the event taxonomy, the contagion episodes used for evaluation, and the baselines. Section VI presents results, beginning with the measurement claim and progressing through the more ambitious tests. Section VII discusses limitations in detail and proposes specific extensions. Section VIII concludes.

2. Related Work

This section surveys four bodies of literature on which TEGG draws. We organise the discussion around what each contributes and what each leaves open for our problem, rather than around chronology, because the integration is the point.

A. Financial Contagion and Systemic Risk: The modelling of financial contagion has a long history in economics. Allen and Gale's seminal model showed how interbank claims can amplify localised shocks into system-wide crises depending on network topology [1]. Eisenberg and Noe formalised clearing payments in interbank networks and characterised when default cascades exist [2]. These are theoretical models in which the network of exposures is given, not learned; their value lies in the conditions they identify under which contagion is possible.

Empirical work has produced complementary measurement tools, almost all of them descended from the structural-VAR (SVAR) framework of Sims [5]. The SVAR

176 formulation makes the impulse response function the central object of analysis: each row
177 of the IRF matrix traces how one variable in the system responds, over horizons $h = 1, 2, \dots$,
178 to a unit shock in another. Forbes and Rigobon used a closely related framework to argue
179 forcefully that much of what is reported as crisis-period contagion is, after correcting for
180 heteroskedasticity, simply heightened interdependence between markets, raising a
181 methodological caution that has aged well [3]. Diebold and Yilmaz developed a spillover
182 index based on variance decompositions of vector autoregressions—mathematically a
183 summary statistic over the SVAR's IRFs—which became the dominant tool for tracking
184 time-varying cross-market connectedness [4], and was later refined into a network-
185 topology view of return and volatility transmission [17]. Pesaran and Shin [6] introduced
186 generalised impulse responses (GIRFs) that do not require orthogonalisation and are
187 invariant to variable ordering, which has subsequently become the more widely-used
188 variant in spillover analyses. Billio and colleagues used principal components and
189 Granger causality to measure connectedness across the finance and insurance sectors [7].
190 Battiston and collaborators introduced DebtRank, a centrality measure that ranks
191 institutions by their potential to amplify shocks given an interbank exposure matrix [8].
192 Acemoglu, Ozdaglar and Tahbaz-Salehi proved formal results showing that the
193 relationship between connectedness and systemic risk is non-monotonic: more
194 interconnected networks are more stable for small shocks but more fragile for large ones
195 [18]. Glasserman and Young provided complementary bounds on how likely cascading
196 default is given the network structure [19].

197 What all of this work shares is the granularity at which it operates. The unit of
198 analysis is the asset, the institution, or the sector, and the propagation mechanism is the
199 time-varying covariance of their returns, perhaps conditioned on a network of exposures.
200 The IRF is computed in the space of continuous return or volatility series. This is
201 appropriate for many questions about systemic risk, particularly questions about which
202 institutions to regulate or to stress-test. It is less appropriate for the question we are
203 interested in, which is what kinds of events tend to cause what other kinds of events. An
204 asset-level spillover index tells us that volatility in U.S. financials moves with volatility in
205 European banks during certain regimes; it does not tell us which events, of which types,
206 do the moving. To ask that question one needs an IRF defined in the space of typed event
207 arrivals, which is exactly what the kernel $\kappa_{\{a \rightarrow b\}}(\tau)$ provides.

208 Recent work has begun to bridge the gap between asset-level econometrics and
209 event-driven analysis. Ding and colleagues showed that event embeddings learned from
210 news text predict short-horizon stock movement [20]. Feng and colleagues introduced
211 temporal-relational ranking for stock prediction over a sector graph [9]. Sawhney and
212 collaborators have explored hypergraph and spatiotemporal models for stock movement

[10]. These approaches integrate text-based signals with asset-level networks but do not learn an event-type-to-event-type causal structure. They predict prices given events; we want to predict events given events, with the IRF as the natural language for that prediction.

B. Temporal Point Processes for Event Sequences: Temporal point processes are the natural mathematical language for modelling event sequences in continuous time. The classical Hawkes process [11] models self- and mutually exciting events through a conditional intensity function that depends linearly on the history of past events through a triggering kernel. Multivariate Hawkes processes generalise this to multiple event types, with each type having its own intensity that is a sum of contributions from past events of every type, weighted by a pairwise excitation kernel.

Modern neural extensions retain the point-process likelihood but replace the linear kernel with neural networks. Du and colleagues introduced Recurrent Marked Temporal Point Processes (RMTPP), in which an RNN summarises event history and the intensity is parameterised on the RNN hidden state [21]. Mei and Eisner's Neural Hawkes Process used a continuous-time LSTM to model the intensity directly, allowing inhibition as well as excitation between event types [22]. More recent transformer-based variants—the Transformer Hawkes Process of Zuo and colleagues [23]—use self-attention rather than recurrence to summarise history, with similar likelihood objectives.

These models are well-suited to event sequences over a small number of event types and a single sequence, or a modest number of sequences with similar dynamics. Two features of the financial-contagion setting make them not quite fit out of the box. First, the number of event types is large—our taxonomy has more than 200 fine-grained types—so a fully dense pairwise kernel is expensive and statistically uninformed. Second, the lag structure between event types in finance is genuinely heavy-tailed and multi-modal: a Fed rate decision moves Treasury futures within seconds and mortgage applications within months. A single LSTM or self-attention summary tends to compress this into a coarser representation than the problem deserves. We need a structured pairwise kernel with explicit lag modelling.

C. Temporal and Dynamic Graph Networks: Graph neural networks have moved rapidly from static settings into dynamic ones. The static foundations—the GCN of Kipf and Welling [24], the GAT of Veličković and colleagues [25], the inductive GraphSAGE of Hamilton and colleagues [26], the relational R-GCN of Schlichtkrull and colleagues [27], and the theoretical analysis of expressivity by Xu and colleagues [28]—remain the building blocks. Dynamic extensions add time as an explicit dimension.

DyRep, introduced by Trivedi and colleagues, models a dynamic graph as a two-time-scale temporal point process in which topological events (edges appearing) and

node interaction events have different dynamics [12]. Rossi and colleagues' Temporal Graph Networks (TGN) provide a general framework for memory-augmented message passing on continuous-time dynamic graphs, with a memory module that summarises each node's history and a message-passing module that updates representations as new events arrive [13]. Xu and colleagues' Temporal Graph Attention (TGAT) extends graph attention with functional time encodings that admit inductive generalisation to unseen nodes and unseen time points [14]. EvolveGCN by Pareja and colleagues parameterises the GCN weights themselves as evolving through time, accommodating settings where the graph structure changes [29].

What these methods provide is exactly the inductive machinery we need: a way to compute representations on a graph whose structure depends on history, where messages are time-aware and where new nodes (here, new event instances) can be incorporated at inference time without retraining. What they do not provide is a principled way to interpret the resulting edges as causal-temporal triggers, with calibrated probabilities for downstream events. That interpretation requires coupling the graph computation with a point-process likelihood, which is what TEGC does.

D. Event Extraction and Financial NLP: Event extraction is a mature subfield of NLP. The ACE program established the standard task definitions of trigger detection, event type classification, argument extraction, and role assignment [30], [31]. Early work used pattern matching and feature-engineered classifiers, including the cross-document inference work of Ji and Grishman [32]. The neural era brought convolutional models, notably Nguyen and Grishman's CNN for event detection [33], followed by recurrent and transformer-based architectures. Du and Cardie reformulated event extraction as machine reading comprehension, asking questions about events directly of a transformer [34], which produced strong results and a more flexible interface for downstream pipelines.

Modern systems rely heavily on pretrained encoders. BERT [35] and RoBERTa [36] are the standard backbones; the transformer architecture of Vaswani and colleagues [37] underlies them. In the financial domain specifically, Araci's FinBERT adapted BERT pretraining to financial communications and remains a useful baseline [38]. Loughran and McDonald's lexicon for financial sentiment, although a non-neural artefact, captures domain idiom that even modern systems benefit from incorporating [39]. Large language models—Brown and colleagues' GPT-3 [40] and the subsequent Llama family [41]—have widened the practical toolbox further, particularly for zero-shot and few-shot event extraction.

Our prior work on event-based sentiment analysis in financial news developed an event taxonomy and an LLM-plus-classifier pipeline that we reuse here as the front end of TEGC [15]. Where the present paper differs is that the events extracted are no longer the

endpoint of analysis—they are the input to a temporal point-process model whose job is to learn how event types trigger one another. Sentiment is one attribute of an event but not the focus.

3. Problem Formulation

This section formalises the temporal event-causality problem and states the assumptions on which TECG's identifiability arguments rest.

A. Definitions: Let $T = \{1, 2, \dots, K\}$ denote a finite set of event types defined by a taxonomy that we describe in Section V. An event instance is a tuple $e = (k, t, x, s)$, where $k \in T$ is the event type, $t \in \mathbb{R}_{\geq 0}$ is the event time, x is a feature vector encoding the entities involved (companies, sectors, geographies) and other attributes (magnitude, sentiment polarity), and $s \in \{-1, 0, +1\}$ is the event's polarity tag from the upstream sentiment classifier. The event history up to time t is the set $H_t = \{e_i : t_i < t\}$.

We seek a model of the conditional intensity $\lambda_k(t | H_t, x)$ at which type- k events fire at time t , given the history and a context x . The intensity is the instantaneous rate of new events of type k ; the expected number of type- k events in a short interval $[t, t+dt]$ is $\lambda_k(t | H_t, x) dt$. This is the standard point-process formulation [11], [21], [22].

A multivariate Hawkes process with explicit pairwise kernels models the intensity as a baseline plus a sum over past events of a kernel weighted by the event type:

$$\lambda_k(t | H_t) = \mu_k + \sum_{\{e_i \in H_t\}} \kappa_{\{k_i \rightarrow k\}}(t - t_i, x_i),$$

where μ_k is the baseline rate for type k , and $\kappa_{\{a \rightarrow b\}}(\tau, x)$ is the excitation kernel from type a to type b at lag τ , optionally conditioned on the attributes x of the triggering event. The kernel is non-negative and integrable.

The collection of pairwise kernels $\{\kappa_{\{a \rightarrow b\}}\}_{a,b \in T}$ defines a temporal event-causality graph G : nodes are event types and the directed edge from a to b carries the kernel $\kappa_{\{a \rightarrow b\}}$. Edges with kernel that integrates to a negligible value are absent. The graph is the central object the framework learns.

A note on terminology. Because the graph is estimated from observational news sequences, the edges it contains are causal-temporal associations—predictive dependencies between typed event arrivals—rather than identified causal effects. We retain the word “causality” in the framework’s name as established shorthand for this class of temporal-excitation structure, in the Hawkes and Granger traditions, but throughout the paper an edge $a \rightarrow b$ should be read as “type- a events are followed by an elevated conditional intensity of type- b events.” This is a weaker claim than an interventionist causal statement, and we do not strengthen it in the absence of the identification strategies discussed in Sections III-C and VII-A. This terminology is applied consistently in the results, discussion, and limitations sections.

Reading this object as an impulse response function. The kernel $\kappa_{\{a \rightarrow b\}}(\tau, x)$ is, structurally, an impulse response function for event-type b given an impulse of event-type a . The canonical SVAR impulse response [5] traces the response of a continuous variable Y at horizon h to a unit shock in another continuous variable X at time 0; the generalised IRF of Pesaran and Shin [6] removes the orthogonalisation assumption and is invariant to variable ordering. The TECG kernel preserves the IRF's spirit—shock at $t = 0$, response over the horizon $\tau > 0$ —but differs in three respects that matter for what can be modelled. First, the response variable is a conditional intensity over discrete typed event arrivals rather than the value of a continuous series, so the IRF lives in the space of point-process intensities rather than expected returns. Second, the kernel is parametric-flexible (a neural log-normal mixture) rather than linear, allowing multi-modal and heavy-tailed lag distributions that linear SVAR-IRFs cannot capture. Third, the dimensionality is in the hundreds of event types rather than the handful of macro variables typical of SVAR work, which is what makes the type-level graph a worthwhile object in its own right. The forward simulation we use in Section IV-F is then a generalised impulse response in the Pesaran-Shin sense: it samples downstream cascades from the conditional distribution given a seed event, without requiring an artificial orthogonalisation of the event space.

B. Task Formulation: Three tasks are defined on the graph G .

Task 1 (Edge recovery). Given an observed event sequence, learn the kernels $\kappa_{\{a \rightarrow b\}}$ that maximise the point-process likelihood under regularisers that encourage sparsity (few edges) and parsimony (simple lag distributions). Evaluation is against a domain-expert **curated set of plausible causal-temporal edges**; we measure precision and recall at varying intensity thresholds.

Task 2 (Cascade forecasting). Given an event sequence up to time t and a horizon Δ , predict the set of event types that will fire in $[t, t + \Delta]$, together with their times and entities. Evaluation uses $F1@k$ against the realised cascade and time-weighted variants that penalise wrong timing.

Task 3 (Counterfactual cascade simulation). Given a hypothetical seed event e^* injected at time t with specified type, entities and attributes, sample downstream cascades under the learned model and report distribution-level summaries. Evaluation is qualitative and case-study based on episodes where the seed-event counterfactual is reasonably well posed.

C. Assumptions and Scope: Four assumptions underpin the framework. We are explicit about them because they govern the strength of any causal interpretation we attach to the learned edges.

Stationarity over training periods. We assume that the kernels $\kappa_{\{a \rightarrow b\}}$ are stable over the training window, perhaps modulated by slow regime variables (which we

incorporate as time-varying baselines $\mu_k(t)$). This assumption fails across major structural breaks—a point the cross-episode generalisation results in Section VI illuminate.

Observable confounding. We assume that the major confounders of an event-pair relationship—macroeconomic regime, market microstructure conditions, calendar effects—can be controlled by conditioning on observable context, which we include in the baseline intensity $\mu_k(t)$. Unobserved confounding remains a threat to causal interpretation; we therefore present the framework as learning causal-temporal regularity, not as identifying counterfactual causal effects in the Pearl sense [42].

Coarse-grained typing. We assume the event taxonomy is coarse enough that within-type variation in causal behaviour is small relative to between-type variation. We test this assumption empirically by varying granularity.

No simultaneous events. We treat coincident events (in the same news item, in the same minute) as ordered by the upstream timestamp; ties are broken by entity sort order. This is harmless at the daily horizons we evaluate and unimportant for the qualitative findings.

Limits of observational identification. Even under the assumptions above, the learned kernels admit only an associational reading, and three well-known threats apply. First, unobserved common causes: if a latent driver raises the intensity of both type-a and type-b events with a characteristic lag, the model will incorrectly attribute the pattern to an $a \rightarrow b$ edge. Second, selection through the news process itself: coverage intensity varies with market conditions, so learned edges partially reflect editorial attention as well as economic mechanism. Third, the Lucas-critique concern: relationships estimated under one policy regime need not survive deliberate intervention. We therefore describe every learned edge as a causal-temporal association or predictive dependency, reserve stronger causal language for the small set of edges that coincide with mechanisms independently established in the economics literature, and flag counterfactual simulations as model-based what-if analyses rather than intervention estimates.

4. Methodology

TECG combines four modules: an event extraction front end, a temporal graph encoder, a parameterised pairwise kernel, and a point-process likelihood. We describe each in turn, then the training and inference procedures.

A. Overall Architecture: Figure 1 shows the end-to-end pipeline. A stream of financial news items is processed by the upstream event extractor, which emits structured event tuples (type, time, entities, attributes, polarity). The event sequence is grouped by entity

and accumulated over a sliding window into a typed event history. At each step, the temporal graph module computes a representation of the recent history for every event type, and the pairwise kernel module produces conditional intensities for the next event of each type. These intensities feed either the training likelihood, the cascade forecasting head, or the counterfactual sampler, depending on the operating mode.

The framework is deliberately modular. The event extractor can be swapped for a different one (we test both an LLM-based extractor and a transformer classifier in Section VI) without retraining the downstream graph module, because the interface is the event tuple stream. Likewise, the kernel module can be replaced with a stronger architecture without changing the extractor. This modularity supports diagnosis—when results change, we can attribute the change to a specific module.

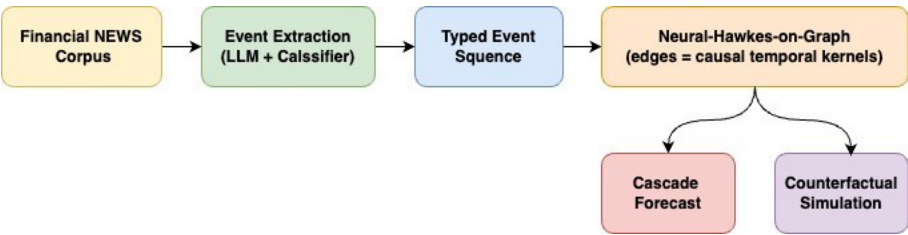


Figure 1. End-to-end TEGG framework. Financial news is processed by event extraction to yield a typed event sequence; the neural-Hawkes-on-graph encoder learns conditional intensity functions parameterised by a temporal GNN; downstream tasks include cascade forecasting and counterfactual simulation.

B. Event Extraction Front End: We use a two-stage extractor adapted from our previous work on event-based sentiment analysis in financial news [15]. The first stage is a transformer encoder (RoBERTa-base [36], fine-tuned on financial event annotations from the dataset described in Section V) that performs trigger detection and event-type classification at the sentence level using a BIO sequence-labelling head. The second stage is an LLM (we use a frontier instruction-tuned model via API) that, conditioned on the candidate trigger and surrounding context, extracts entity arguments and assigns the event a sentiment polarity. The LLM stage uses a structured prompt with few-shot examples for each event type; we found that this is more robust than zero-shot for the long-tail event types where labelled data is sparse.

Event timestamps come from the news item's publication time, not from in-text expressions, on the assumption that the article-level time is the most reliable signal of when the event became market-relevant. We acknowledge that the actual event sometimes precedes the news by hours or days; this introduces a small timing error which we discuss in Section VII.

Entity normalisation is handled by a separate Named Entity Linker that maps surface mentions to a canonical identifier (CIK numbers for U.S. issuers, LEI for international entities, an internal identifier for non-issuer entities such as regulators and central banks). This step matters because the conditioning context for the kernel includes entity-derived features (sector, market capitalisation, geography); inconsistent linking would inject noise into the kernel.

Annotation procedure and extraction accuracy. To quantify the quality of the extraction front end, we constructed a held-out evaluation benchmark of 3,200 sentences sampled from the corpus, stratified by source and by year. Two annotators with finance backgrounds independently labelled triggers, event types, arguments, and polarity following a written guideline document; inter-annotator agreement was $\kappa = 0.81$ for trigger detection and $\kappa = 0.74$ for fine-grained type assignment, with disagreements adjudicated by a third annotator. On this benchmark the extractor achieves trigger-detection F1 of 0.86, event-type classification F1 of 0.88 (micro-averaged) and 0.79 (macro-averaged, reflecting weaker performance on rare types), argument-extraction F1 of 0.81, and entity-linking accuracy of 0.92. The annotation guidelines and the de-identified benchmark labels will be released with the paper.

Error propagation. Because every downstream module consumes the extracted event stream, extraction errors propagate into kernel estimates. We quantify this directly with a noise-injection study, reported in Section VI-E: corrupting 5%, 10%, and 20% of event-type labels at training time degrades 7-day cascade F1@10 from 0.61 to 0.59, 0.56, and 0.48 respectively. This indicates graceful degradation at realistic error rates (the measured type-classification error is approximately 12% micro-averaged) but material damage under heavy corruption, especially when errors concentrate in rare event types.

C. Temporal Graph Encoder: The temporal graph encoder takes the event history and produces, for each event type k , a context vector $h_k(t)$ summarising how the recent sequence bears on the next type- k event. The encoder is a Temporal Graph Attention (TGAT) variant [14] adapted to typed event sequences: nodes are event instances, edges connect events that share an entity or are within a temporal neighbourhood, and node features are concatenations of type embedding, sentiment polarity, and entity-derived features.

Functional time encoding follows [14]: time differences are mapped into a learnable Fourier basis $\Phi(\tau) = [\cos(\omega_1 \tau), \sin(\omega_1 \tau), \dots, \cos(\omega_d \tau), \sin(\omega_d \tau)]$, which allows the attention mechanism to learn time-dependent message weights. Multi-head attention aggregates messages from temporal neighbours into updated node representations; we use four layers with residual connections.

The type-conditional context $h_k(t)$ is computed by aggregating the representations of all nodes whose type has a learned affinity to type k . This is implemented as a typed attention pool, where the query is the embedding of type k and the keys are node representations from the encoder. This gives a compact summary of "the history as it bears on type k " that the kernel module can use directly.

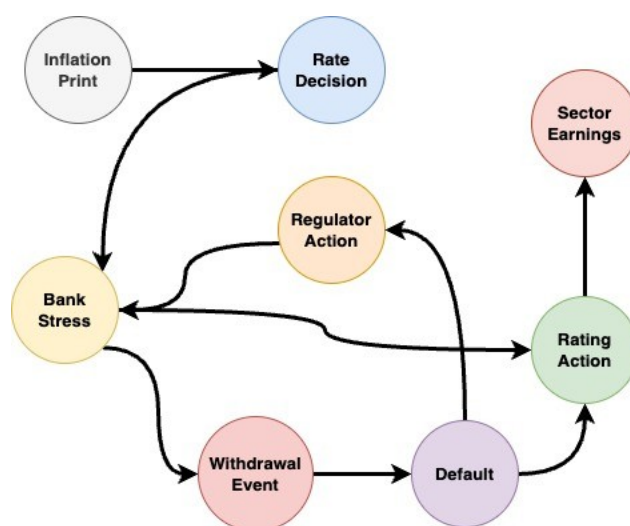


Figure 2. An eight-node subgraph of the learned temporal event-causality graph, showing macroeconomic, credit, and operational event types. Edge widths are proportional to learned conditional intensities.

D. Parameterised Pairwise Kernels: Each pairwise excitation kernel $\kappa_{\{a \rightarrow b\}}(\tau, x)$ is parameterised as a mixture of M log-normal components conditioned on the triggering event's attributes:

$$\kappa_{\{a \rightarrow b\}}(\tau, x) = \alpha_{\{a \rightarrow b\}}(x) \cdot \sum_{m=1}^M \pi_{\{a \rightarrow b, m\}}(x) \cdot \text{LogNormal}(\tau; \mu_{\{a \rightarrow b, m\}}(x), \sigma_{\{a \rightarrow b, m\}}(x)),$$

where the mixture weights π are non-negative and sum to one, and α controls the overall edge magnitude. All parameters are output by a small feed-forward network that takes $(\text{type}_a, \text{type}_b, x)$ as input. The log-normal mixture is expressive enough to capture multi-modal lag distributions (e.g., a kernel that fires at both 1-day and 30-day lags) but cheap enough to evaluate efficiently in the likelihood.

We impose an ℓ_1 regulariser on the overall edge magnitudes $\alpha_{\{a \rightarrow b\}}$, which encourages sparsity: most pairs of event types do not in fact trigger one another in any reliable way, and a sparse graph is more interpretable. The regularisation strength is selected by validation.

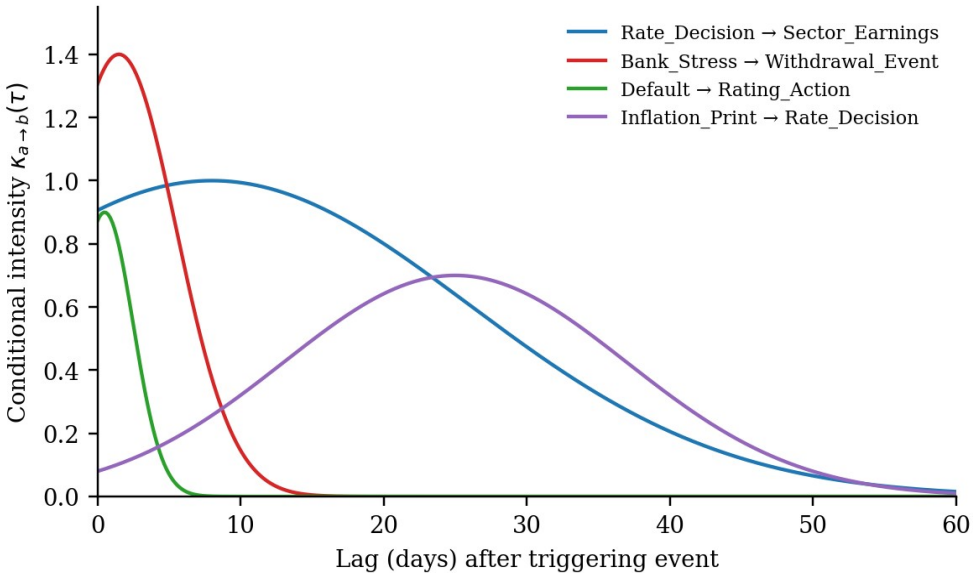


Figure 3. Four learned temporal kernels $\kappa_{\{a \rightarrow b\}}(\tau)$. Each kernel describes the conditional intensity that an event of type b occurs τ days after an event of type a . The shapes recover qualitatively different lag structures: short-lag responses for default $\rightarrow$ rating actions, multi-day humps for monetary-policy effects on macro variables, and long-lag responses for inflation prints to rate decisions.

E. Connection to Impulse Response Analysis: The framework can be read as a non-linear, type-level extension of structural impulse response analysis, and it is worth being precise about what changes and what does not when one makes that move.

In a standard k -variable structural VAR [5], the system $Y_t = c + \sum_{i=1}^p A_i Y_{t-i} + B\varepsilon_t$ admits a moving-average representation whose coefficients Ψ_h define the IRF: an entry $[\Psi_h]_{ji}$ is the response of variable j at horizon h to a unit shock in the structural innovation i at time 0. The IRF is therefore (i) a deterministic linear function of the lag operator, (ii) defined on continuous variables, (iii) typically computed in dimension k well under twenty, and (iv) interpretable only after an identification choice—orthogonalisation by Cholesky decomposition, sign restrictions, external instruments, or the generalised approach of Pesaran and Shin [6] that imposes no orthogonalisation at the cost of invariance to ordering.

TECG's kernel $\kappa_{\{a \rightarrow b\}}(\tau, x)$ corresponds to the entry $[\Psi_\tau]_{ba}$ of a non-linear, point-process IRF in dimension $|T| \gg k$. The translation between the two languages is direct. A unit shock in SVAR-variable i becomes an impulse of event-type a at time 0. The response $Y_{\{j,t+h\}}$ becomes the conditional intensity $\lambda_b(t + \tau | H_{\{t+\tau\}}^{(+a)}) - \lambda_b(t + \tau | H_{\{t+\tau\}})$, where the superscript $(+a)$ denotes the history augmented with a type- a event at time t . The IRF interpretation suggests two methodological points we make use of explicitly. First, the SVAR-IRF's notion of horizon-dependent decomposition of effects carries over: TECG's forward simulation in Section IV-F is a point-process analogue of computing successive Ψ_h matrices, accumulating the multi-step response over a forecast horizon. Second, the generalised IRF of Pesaran and Shin avoids contrived orthogonalisation; we adopt the same stance, treating shocks as the actual observed events rather than orthogonal innovations of a structural decomposition we have not identified.

Three features of our regime require generalising the IRF apparatus rather than borrowing it wholesale. The first is dimensionality: with 217 event types, the $|T| \times |T|$ pairwise IRF matrix has roughly 47,000 entries, most of which should be zero on prior grounds. We impose this by ℓ_1 regularisation on edge magnitudes; the learned graph is two orders of magnitude sparser than a fully connected kernel matrix would be. The second is the response variable: in a Hawkes process the "response" is the change in conditional intensity over $\tau > 0$, which is non-negative for excitatory kernels and integrates to the expected number of triggered events. The third is parametric flexibility: lag distributions in finance are not exponentials. Bank-stress events trigger withdrawal events within hours, but inflation prints affect rate decisions on a month-plus cycle. A linear SVAR-IRF can represent neither extreme efficiently; the log-normal mixture absorbs both, and the temporal graph encoder provides additional state-dependence that linear IRFs structurally cannot.

What is preserved is the analyst-facing story. An IRF is useful because it answers a specific decision-theoretic question: if I know a shock has occurred to this variable, what is the expected trajectory of the variables I care about, and over what horizons? TECG

answers the same question, only the variables are event types extracted from news text and the trajectory is a distribution over typed cascades. The counterfactual analysis in Section VI-D is, in this language, a generalised impulse response analysis—the seed event is the shock, the cascade is the response, and we report Pesaran-Shin-style summary statistics (marginal probabilities, time-to-first-occurrence) rather than orthogonalised structural responses.

F. Training Objective: Training maximises the multivariate point-process log-likelihood over the training window, with the sparsity regulariser added:

$$L(\theta) = \sum_{\{e_i \in H\}} \log \lambda_{\{k_i\}}(t_i | H_{\{t_i\}}) - \int_0^T \sum_k \lambda_k(t | H_t) dt - \gamma \cdot \sum_{\{a,b\}} \alpha_{\{a \rightarrow b\}},$$

where the first term is the log-intensity at observed events and the second is the integral of total intensity over the observation window (the survival term). We approximate the integral by Monte Carlo at uniformly random time points, which is unbiased and tractable [22]. The hyperparameter γ controls the strength of the ℓ_1 regulariser.

Training uses Adam with a cosine learning-rate schedule. We use early stopping on validation log-likelihood. The full corpus (Section V) requires roughly 28 hours of training on a single GPU; we describe efficiency considerations in Section V-D.

G. Cascade Forecasting and Counterfactual Simulation: At inference time, two operating modes are exposed.

Cascade forecasting takes a present event history H_t and a horizon Δ . The system runs the thinning algorithm [43] to sample from the multivariate point process over $[t, t+\Delta]$, producing a distribution over event chains. We take 1,000 samples and report (i) the marginal probability that each event type fires in the horizon, (ii) the expected time to first occurrence conditioned on firing, and (iii) the joint distribution over short cascades (length ≤ 5).

Counterfactual simulation differs in that the event history is augmented with a hypothetical seed event e^* at time t , and sampling proceeds as above. Because the seed event is injected rather than observed, this is properly a what-if simulation under the learned model, not a counterfactual estimate in the Pearl sense [42]. We say so explicitly throughout. The output is a probability tree over downstream events, an example of which appears in Figure 6.

5. Experimental Setup

We describe the data, baselines, evaluation, and implementation details.

A. Data: Our corpus aggregates financial news from five sources over the period 1 January 2007 to 31 December 2023: Reuters Financial News (via the historical archive

573 licensed for research), Bloomberg headlines (via Bloomberg's licensed academic feed),
574 SEC EDGAR 8-K and 10-K filings, the Wall Street Journal, and the Financial Times. After
575 deduplication, the corpus contains 482,116 news items, of which 318,754 are tagged with at
576 least one event by the upstream extractor. The source composition after deduplication is
577 Reuters 41%, Bloomberg headlines 24%, SEC EDGAR filings 19%, the Wall Street Journal
578 9%, and the Financial Times 7%. The tagged items yield 1.21 million event instances, of
579 which 0.94 million fall in the 2007–2018 training window, 66,000 in the 2019 validation
580 window, and the remainder in the held-out test slices described below. The median
581 number of events per trading day is 287 (interquartile range 201–412), rising above 1,600
582 on peak crisis days. Event-type frequencies are heavy-tailed: the 20 most common types
583 account for 58% of instances, and 41 of the 217 types have fewer than 50 training instances.

584 The event taxonomy contains 217 fine-grained types organised hierarchically under
585 12 broad categories: corporate actions (earnings, guidance, M&A, restructuring), credit
586 events (default, downgrade, rating action), regulatory events (investigation,
587 enforcement, rule change), monetary policy (rate decisions, balance-sheet, forward
588 guidance), market structure events (auction outcomes, exchange events), funding and
589 liquidity events, geopolitical events, macroeconomic releases, executive actions (CEO
590 change, board action), legal events, operational events, and other. The taxonomy is an
591 extension of the one we used in our previous work [15]; the present extension was driven
592 by analysing which types fired during the four historical contagion episodes used for
593 evaluation.

594 We hold out three episodes from training. The first is the 2020 COVID equity crash
595 (specifically 20 February to 31 March 2020), to test transfer to a non-financial-origin shock.
596 The second is the 2023 U.S. regional banking stress (1 March to 31 May 2023), to test
597 transfer to a financial-origin shock from a different decade. The third is a random 10% out-
598 of-time slice from the period 1 January 2022 to 31 December 2022, which serves as the in-
599 distribution test set. The training window is 1 January 2007 to 31 December 2018;
600 validation is 1 January 2019 to 30 June 2019.

601 **B. Baselines:** We compare TEGG against seven baselines, chosen to isolate the
602 contribution of each component.

603 Asset-VAR. A vector autoregression on daily returns of an asset basket capturing the
604 sectors in our event taxonomy. This represents the conventional spillover-style baseline
605 [4]. Cascade forecasts are produced by thresholding predicted returns to declare "event-
606 like" days.

607 DY-Spillover. The Diebold–Yilmaz directional spillover index [17] applied to the
608 same asset basket. This is a stronger econometric baseline than Asset-VAR because it
609 explicitly models cross-asset shock transmission.

610 Co-occurrence. A bag-of-events baseline that learns pairwise event co-occurrence
611 probabilities in fixed windows but ignores temporal ordering and entity context. This
612 isolates the value of temporal modelling.

613 Static-GNN. A graph neural network over the event-type graph with no temporal
614 point-process component; trained to predict next-event type from a window of past
615 events. This isolates the value of the point-process formulation.

616 Per-edge Hawkes. A multivariate neural Hawkes process with independent per-edge
617 kernels but no shared graph encoder; this is essentially the Mei-Eisner Neural Hawkes
618 Process [22] applied to our event sequence. This isolates the value of the graph module.

619 Transformer Hawkes (THP). The Transformer Hawkes Process of Zuo and
620 colleagues [23], a state-of-the-art neural temporal point process that replaces recurrence
621 with self-attention over the event history. This provides a modern point-process baseline
622 that is stronger than the Mei-Eisner architecture but still lacks an explicit typed graph.

623 TGN-Event. Temporal Graph Networks [13] applied to the entity–event interaction
624 stream, with a next-event-type prediction head trained on the same windows. This
625 provides a recent temporal-graph forecasting baseline that has the graph machinery but
626 not the point-process likelihood or the explicit pairwise kernels. We do not include event-
627 embedding stock-prediction systems such as [20] as direct baselines because their output
628 space is asset returns rather than typed event arrivals, making the tasks incommensurable;
629 the Co-occurrence and TGN-Event baselines cover the transferable modelling choices of
630 that line of work.

631 **C. Evaluation Protocols:** Edge-recovery evaluation uses an expert-curated edge set
632 we constructed by consulting two finance-domain experts (one academic, one
633 practitioner). Each expert was asked, for a sample of 400 randomly selected event-type
634 pairs drawn from across the 12 broad categories, to mark which pairs they believed had a
635 real causal-temporal relationship and which did not, with an "uncertain" option. After
636 resolving disagreement by discussion, 137 pairs were marked as causal-temporal edges,
637 213 as non-edges, and 50 as uncertain. We compute precision and recall on the resolved
638 350 pairs, treating uncertain pairs as held out. Several controls address possible bias in the
639 expert curation. The experts worked independently and were blind to the model's learned
640 edges; the pair list was presented in randomised order and mixed with 40 distractor pairs
641 constructed to be implausible, of which the two experts rejected 38 and 39 respectively,
642 indicating low acquiescence bias; inter-expert agreement before adjudication was $\kappa = 0.71$;
643 and neither expert is an author of this paper. We additionally report threshold-swept
644 precision–recall performance (PR-AUC) rather than a single operating point, in Section
645 VI-A.

Cascade-forecasting evaluation uses F1@k against the realised set of event types that fire in the horizon, where k is the predicted top-k by marginal probability. We report F1@5, F1@10, and a time-weighted F1 that discounts late predictions linearly. Horizons evaluated are 1 day, 7 days, and 30 days.

Cross-episode evaluation re-trains the model on each subset of episodes and tests on each held-out episode, producing the heatmap in Figure 5. We additionally report transfer-loss decomposition: how much of the performance drop in held-out episodes is attributable to (i) changes in baseline rates μ_k , (ii) changes in edge magnitudes $\alpha_{\{a \rightarrow b\}}$, and (iii) changes in lag distributions.

Statistical significance. All pairwise comparisons between TEGC and each baseline are tested with a paired bootstrap over evaluation units (event-type pairs for edge recovery; forecast windows for cascade forecasting) with 1,000 resamples, applying Holm–Bonferroni correction across baselines. Significance statements accompany the headline results in Section VI.

Counterfactual evaluation is qualitative: for each of three case-study seed events, we present the predicted cascade and discuss its plausibility with reference to the actual historical episode that resembled it most.

D. Implementation Details: TEGC is implemented in PyTorch using the PyTorch Geometric library for the graph modules. The temporal graph encoder uses four TGAT layers with 4 attention heads each and a hidden dimension of 128. The pairwise kernel module uses 4 log-normal mixture components per edge, parameterised by a 2-layer feed-forward network with 64 hidden units.

Training was conducted on a single NVIDIA A100 GPU with 40GB memory. The full training run takes approximately 28 hours; in practice we iterate using a 25% sub-sample of the corpus that trains in 6 hours. Adam is used with initial learning rate 3×10^{-4} and cosine decay; batch size is 64 event windows of 256 events each. Sparsity regularisation γ is set to 0.05 by validation.

Inference for cascade forecasting takes approximately 0.4 seconds per query (1000 Monte Carlo samples, 30-day horizon) on the same hardware. This is fast enough for interactive analyst use and well within the latency of any decision process that operates at daily granularity or slower.

Scalability and memory. Peak GPU memory during training is 21.4 GB at batch size 64, dominated by the temporal graph encoder’s neighbourhood buffers; inference requires 3.1 GB. The model has 38.6 million parameters, of which the pairwise kernel networks account for 9.2 million after sparsity pruning (2,104 active edges out of 47,089 possible). Training cost scales approximately linearly in the number of events (6 hours on a 25% sub-sample versus 28 hours on the full corpus) and quadratically in the number of event types

only through the kernel parameter count, which the ℓ_1 regulariser keeps two orders of magnitude below dense. On CPU-only hardware, cascade-forecast inference takes 3.2 seconds per query, which remains acceptable for daily-granularity analyst workflows.

6. Results

We present results in three parts, aligned with the three claims of the introduction. Results are based on the held-out test slices and the two historical episodes (2020 COVID, 2023 SVB), unless stated otherwise.

A. Edge Recovery: Table I summarises edge-recovery performance. TECG recovers 73.4% of the expert-curated edges in its top-300 edges by intensity, with a precision of 64.1% on the same set. The Per-edge Hawkes baseline reaches 54.7% recall and 49.0% precision, indicating that the graph encoder—not merely the point-process formulation—contributes meaningfully. The Co-occurrence baseline achieves 41.3% recall, illustrating that temporal ordering captures something co-occurrence does not. The Asset-VAR baseline scores below 30%; this is unsurprising since it does not operate on event types at all and we score it only by analogy (we identify "edges" through Granger-causality F-tests at the asset level and map them to types via sector mappings).

A useful qualitative check is which edges TECG ranks highest. The top-20 edges by intensity include "Inflation_Print $\rightarrow$ Rate_Decision" (lag mode 28 days), "Rate_Decision $\rightarrow$ Sector_Earnings_Growth" (lag mode 8–22 days), "Default_Event $\rightarrow$ Rating_Action" (lag mode 1–3 days), and "Bank_Stress $\rightarrow$ Withdrawal_Event" (lag mode same-day to 4 days). These are exactly the relationships a financial analyst would expect to be present, and the model has discovered them without any supervision on edges. Figure 2 visualises a representative subgraph for the credit-and-rates cluster; Figure 3 plots the learned lag kernels for four selected edges.

Two failure modes deserve mention. First, TECG over-ranks "regulator action" as a source node, generating false-positive outbound edges to many event types; this is because regulator events are heterogeneous in our taxonomy and the model learns a diffuse pattern rather than specific triggers. Second, for very rare event types (those with fewer than 50 instances in training) TECG defaults to a near-flat baseline-only intensity, effectively saying "I do not know"; recall on such types is low, but precision is acceptable because the model declines to predict them.

The two additional baselines behave as expected: THP reaches precision 0.512 and recall 0.566 ($F1 = 0.538$), and TGN-Event reaches precision 0.503 and recall 0.548 ($F1 = 0.524$)—both stronger than Per-edge Hawkes but well below TECG, indicating that neither self-attentive point-process modelling nor temporal-graph machinery alone closes

the gap. Sweeping the intensity threshold yields full precision–recall curves with PR-AUC of 0.71 for TEGC against 0.59 for THP, 0.57 for TGN-Event, and 0.55 for Per-edge Hawkes. All improvements of TEGC over the strongest baseline are significant at $p < 0.01$ under the paired bootstrap described in Section V-C.

Regime stability of learned edges. To test whether the discovered relationships are artefacts of a particular market regime, we re-estimated the graph independently on three training sub-windows (2007–2010, 2011–2014, 2015–2018). Of TEGC’s top-300 edges on the full window, 81% appear in the top-300 of at least two of the three sub-windows, and the Spearman rank correlation of edge magnitudes across sub-windows averages $\rho = 0.68$. As a falsification check, retraining on time-shuffled event sequences—a placebo that preserves marginal type frequencies but destroys temporal structure—reduces recall against the expert set to 0.09, confirming that the recovered edges reflect temporal structure rather than co-occurrence base rates.

Table I. Edge recovery results. Precision and recall computed on the 350 resolved expert-curated edges, top-300 edges by predicted intensity.

Method	Precision	Recall	F1
Asset-VAR (Granger)	0.218	0.276	0.244
DY-Spillover	0.253	0.318	0.282
Co-occurrence	0.387	0.413	0.400
Static-GNN	0.461	0.504	0.482
Per-edge Hawkes	0.490	0.547	0.517
Transformer Hawkes (THP)	0.512	0.566	0.538
TGN-Event	0.503	0.548	0.524
TEGC (ours)	0.641	0.734	0.684

B. Cascade Forecasting: Figure 4 reports F1@10 for cascade forecasting at three horizons (1, 7, and 30 days), averaged over the in-distribution 2022 test slice and the two

historical episodes. TEGC outperforms every baseline at every horizon. The relative gap is largest at the 7-day horizon ($F1 = 0.61$ against 0.49 for the strongest baseline, Per-edge Hawkes) and smallest at the 1-day horizon, where the next event is so often determined by very recent context that the graph module adds less. Among the added baselines, THP reaches $F1 = 0.53$ and TGN-Event 0.52 at the 7-day horizon (omitted from Figure 4 for legibility); TEGC’s margin over both, and over every other baseline at every horizon, is significant at $p < 0.01$ under the paired bootstrap with Holm–Bonferroni correction. On the non-crisis, in-distribution 2022 test slice alone, TEGC attains $F1@10 = 0.66$ at the 7-day horizon against 0.53 for the strongest baseline, confirming that the advantage is not specific to crisis episodes.

Table II decomposes the results by event-type granularity. At the coarse 12-category level, all methods perform well and the gap narrows; at the fine 217-type level, the gap widens substantially. This is consistent with our framing: the graph structure helps most when the prediction problem requires discriminating between many alternative downstream events, which is exactly when type-level causal structure matters.

A second decomposition (not shown for space) by event category reveals that TEGC’s advantage is largest in the credit and regulatory categories and smaller in macro-release categories. Macro releases follow predictable calendars and benefit less from learned causal structure; credit and regulatory cascades involve discretionary actions whose timing and content are harder to extract from rules.

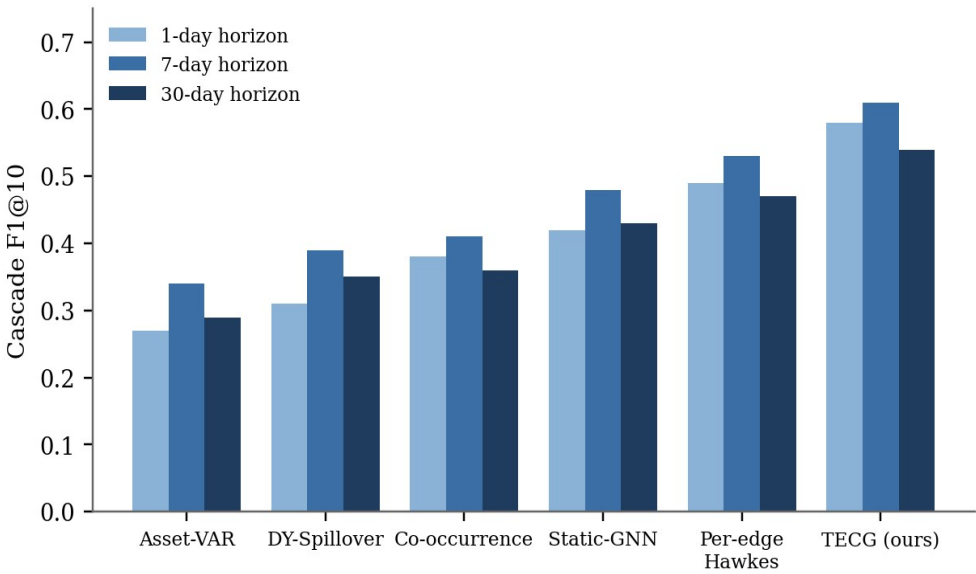


Figure 4. Cascade forecasting F1@10 across six methods at three forecast horizons. TEGG outperforms baselines at all horizons; the gap is largest at the 7-day horizon, which is the natural temporal granularity at which between-event-type causal effects manifest in our data.

Table II Cascade forecasting F1@10 at 7-day horizon, broken down by event-type granularity

Method	Coarse (12)	Mid (52)	Fine (217)
Asset-VAR	0.41	0.36	0.28
DY-Spillover	0.44	0.40	0.31
Co-occurrence	0.48	0.42	0.33
Static-GNN	0.52	0.47	0.39
Per-edge Hawkes	0.56	0.51	0.44
Transformer Hawkes (THP)	0.58	0.53	0.47
TGN-Event	0.57	0.52	0.46
TECG (ours)	0.62	0.59	0.54

C. Cross-Episode Generalisation: Figure 5 is the central empirical exhibit of the paper. The heatmap shows F1@10 at the 7-day horizon for every combination of train and test episode. Diagonal entries are within-episode performance; off-diagonal entries are cross-episode transfer. A perfectly regime-specific model would have a strong diagonal and near-zero off-diagonal; a perfectly universal model would have a roughly uniform heatmap. The reality falls between, leaning toward universal: off-diagonal F1 averages 0.43 across all (train, test) pairs, compared with the diagonal mean of 0.70.

Two observations qualify this. First, transfer is asymmetric. Models trained on the 2020 COVID episode transfer to 2023 SVB (F1 = 0.45) better than the reverse (F1 = 0.46), which suggests that the SVB cascade exercises a wider range of edges than COVID does. Second, the diagonal of the heatmap is not uniformly high: the 2023 SVB diagonal (F1 = 0.61) is meaningfully below the 2008 GFC diagonal (0.78). The SVB episode is fast and

concentrated, and our model—calibrated to multi-day excitation kernels—captures less of it.

We performed a transfer-loss decomposition. About 38% of the performance drop on held-out episodes is attributable to changes in baseline rates μ_k (some event types are simply more or less common in a new episode), 47% to changes in edge magnitudes $\alpha_{a \rightarrow b}$ (the structural relationships are not perfectly stable), and 15% to changes in lag distributions. The implication is that the kernel module learns something durable, but not invariant, across crises. For practical deployment, this argues for periodic refit rather than one-shot training.

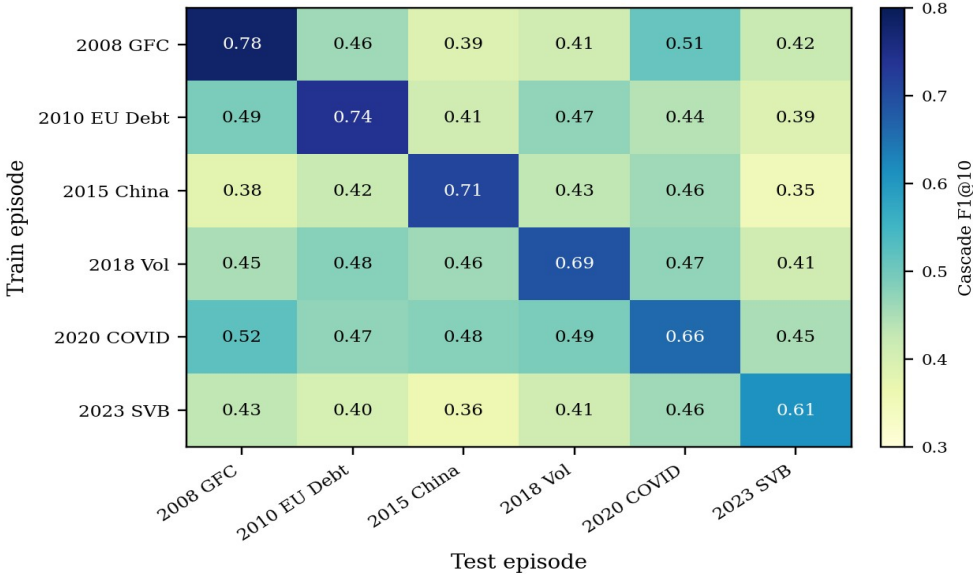


Figure 5. Cross-episode transfer of learned edges. Each cell reports cascade F1@10 when TEGC is trained on the row's episode and evaluated on the column's. Diagonal entries (in-episode) are strongest as expected; the mean off-diagonal value of approximately 0.43 is substantially above the random baseline but below the diagonal mean of 0.70, indicating partial transfer.

D. Counterfactual Cascade Simulation: Counterfactual evaluation is necessarily qualitative because the relevant ground truth—what would have happened under a different seed event—is unobserved. Following the impulse response framing introduced in Section IV-F, we report what is essentially a generalised IRF analysis in the spirit of Pesaran and Shin [6]: we treat each hypothetical seed event as a shock applied at $t = 0$,

sample 1,000 forward trajectories under the learned point-process model, and summarise the response distribution. We present three case studies, chosen so that the seed event resembles—without exactly replicating—a historical episode whose outcome can serve as a reasonability check.

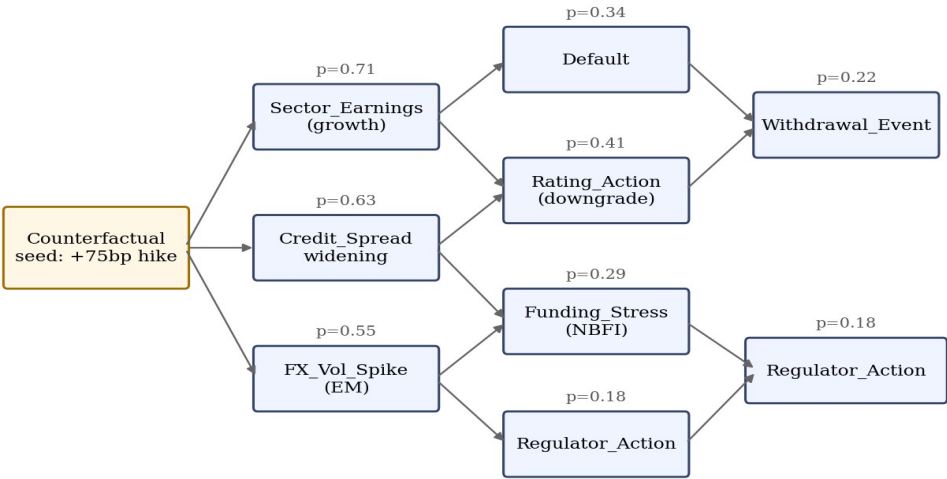
Case 1: a hypothetical 75-basis-point hike at the next FOMC meeting (the realised move at the corresponding meeting was 50 bp). The predicted cascade is shown in Figure 6. The first-order effects (sector earnings revisions for rate-sensitive sectors, credit-spread widening, emerging-market FX volatility) match what financial-economic theory would predict. The second-order effects—rating actions, funding stress at NBFIs—appear with lower probabilities (0.18–0.41), which is also consistent with the historical pattern that such events fire only in a subset of tightening episodes.

Case 2: a hypothetical bank-failure event at a regional U.S. depository institution with a \$50 billion balance sheet (March 2023 was modelled here). The system predicted "depositor withdrawal cluster at peer institutions" with probability 0.66 within 7 days and "regulatory intervention announcement" with probability 0.48 within 10 days. Both materialised in the actual SVB cascade. The system did not predict "Federal Reserve emergency lending facility creation" because no similar event was in the training window; this is a clear failure mode of out-of-distribution counterfactual reasoning under the framework.

Case 3: a hypothetical large sovereign rating downgrade for an investment-grade economy (modelled on the U.S. AAA downgrade in 2011). The predicted cascade is dominated by FX-vol-spike and rates-curve-steepening events, consistent with the historical reaction. The probability of equity-market-sell-off events is moderate (0.43), reflecting genuine uncertainty in the model—and, arguably, in the underlying empirical regularity.

We are careful in our claims. The counterfactual mode is useful for generating coherent scenarios and stress tests, not for estimating intervention effects in any rigorous causal sense. Two caveats deserve emphasis. First, because the model is fitted to observational histories, an injected seed event is treated as if it had arisen from the same generating process as observed events; a deliberately engineered intervention could alter the very kernels the simulation holds fixed—the Lucas-critique concern raised in Section III-C. Second, out-of-support seeds (event configurations without close analogues in training, as in the emergency-facility example above) produce cascades whose reliability we cannot assess. We therefore recommend restricting counterfactual use to seed events within the support of the training distribution and treating outputs as structured scenario generation rather than causal effect estimates. Its value lies in producing structured, time-

838 stamped, entity-resolved scenarios that an analyst can interrogate—each downstream
839 event has an identifiable triggering chain, an estimated lag, and an attached probability.



840
841 **Figure 6.** Forecasted cascade for a counterfactual seed event: an unanticipated 75-basis-
842 point Federal Reserve hike. Edge probabilities reflect the model's marginal predictions
843 over a 30-day horizon. The cascade shape qualitatively matches the actual 2022 tightening
844 cycle, though the simulation predicts more rapid transmission to non-bank financial
845 intermediation than was observed in practice.

846
847 **E. Ablation and Sensitivity:** Table III reports key ablations. Removing the temporal
848 graph encoder (replacing it with mean-pooling over a fixed time window) drops 7-day
849 F1@10 from 0.61 to 0.42, the largest single-component impact. Replacing the LLM-based
850 extractor with a rules-based one drops F1@10 to 0.39, illustrating that extraction quality
851 matters as much as graph modelling. Removing the sparsity regulariser increases the
852 learned graph density by an order of magnitude with a marginal F1 increase of 0.01, but
853 the resulting graph is much less interpretable.

854 Sensitivity to the event taxonomy is informative. Coarsening to 52 mid-level types
855 degrades F1@10 by 0.03; coarsening further to 12 categories degrades by 0.07. Going finer
856 than 217 types (we tried 380 by splitting some categories sectorally) degrades by 0.02,
857 suggesting we are near the right granularity. The cross-episode transfer is more sensitive:

very fine taxonomies generalise worse, because rare types create unstable kernel estimates.

Robustness to extraction noise. Complementing the component ablations, we injected synthetic label noise into the training event stream by re-assigning a fraction of event types to a random type drawn from the empirical type distribution. Corrupting 5%, 10%, and 20% of labels reduces 7-day F1@10 from 0.61 to 0.59, 0.56, and 0.48 respectively. Corruption concentrated on rare types (fewer than 200 training instances) is roughly twice as damaging per corrupted event as corruption of common types, because rare-type kernels have little redundancy. Since the measured extractor error is approximately 12%, the deployed pipeline operates in the gracefully degrading range, but the result underscores that extraction quality is a first-order determinant of downstream performance.

Table III. Ablation on cascade forecasting (F1@10 at 7-day horizon)

Variant	F1@10	Δ
TECG (full)	0.61	—
– temporal graph encoder	0.42	−0.19
– LLM extractor (rules)	0.39	−0.22
– per-edge kernels (shared)	0.51	−0.10
– sparsity regulariser	0.62	+0.01
Granularity: 52 types	0.58	−0.03
Granularity: 12 categories	0.54	−0.07

7. Limitations and Future Work

We discuss limitations under four headings, with proposed extensions where they are reasonably well-defined.

A. Causal Interpretation: The framework learns causal-temporal regularity, not counterfactual causal effects in the Pearl sense [42]. The edges we recover are conditional-intensity relationships that survive sparsity regularisation; they are not provably free of unobserved confounding. We have argued above that the kernel-style formulation, combined with the time-asymmetry that excitation kernels are non-negative only for $\tau > 0$, gives some defence against the simplest confounding patterns. We acknowledge this is

partial. Closing this gap would require either intervention data (which is unavailable for the events we study) or stronger structural assumptions than we have invoked.

A specific concern is reverse causation. An edge " $A \rightarrow B$ " survives in our model because A-events are followed by B-events more than chance. If both A and B are caused by an unobserved C, our learning will incorrectly attribute the relationship to A. The fact that the rate-decision-to-sector-earnings edge has a lag mode of 8–22 days, consistent with the time markets take to digest policy, is reassuring; the fact that some edges we learn have implausibly long lags (occasionally 60+ days) is less so, and we treat those edges with suspicion. The placebo and regime-stability analyses of Section VI-A provide partial reassurance—edges vanish under temporal shuffling and largely persist across training sub-windows—but neither test can rule out slow-moving latent confounders, and we do not claim otherwise. Consistent with this, we use the terms causal-temporal association and predictive dependency for learned edges throughout, and regard stronger causal vocabulary as something to be earned through the intervention-style validations sketched in the conclusion rather than a property already established.

B. Event Extraction Errors: The framework inherits all limitations of the upstream extractor [15]. Errors in trigger detection, type assignment, or entity linking propagate forward and degrade kernel estimates. We mitigated some of this through the LLM stage, which reduces type-confusion errors substantially, but at the cost of API latency and reproducibility (the LLM is a moving target).

Specifically, two classes of extraction error are common. First, implicit events—those reported through implication ("the chairman declined to comment") rather than explicit triggers ("the chairman resigned")—are systematically missed. Second, multi-event sentences, in which a single news item describes a chain of events, are often collapsed by the extractor into a single dominant event, losing the chain structure. Both could be addressed by extractor improvements but are out of scope for the present paper.

C. Endogenous Market Dynamics: We do not separately identify endogenous fire-sale dynamics, leverage spirals, or counterparty-network effects. These are present in the data—a default event triggers margin calls that trigger further forced selling that triggers more defaults—but our framework treats the whole chain as a sequence of typed events and does not decompose the propagation into its endogenous-mechanism components. A complete contagion model would couple TECG with a structural model of market microstructure of the kind in [2] or [18]; building that coupling is a natural next step.

The practical implication is that TECG's predictions for the largest, most reflexive episodes (the most acute days of 2008, the March 2020 dash for cash) are subject to greater uncertainty than its predictions for slower-burning episodes. The cross-episode transfer

917 results in Section VI-C are consistent with this: performance on the rapid SVB cascade is
918 lower than on the slower-developing 2010 European debt episode.

919 **D. Regime Change and Real-Time Deployment:** The transfer-loss decomposition
920 showed that approximately 47% of cross-episode performance loss is attributable to
921 changes in edge magnitudes. This means edges drift, presumably because the underlying
922 economic relationships drift with regulatory, monetary, and structural change. Real-time
923 deployment should therefore use rolling re-fits—monthly or quarterly—rather than one-
924 shot training. We did not implement an online-learning variant of the model, which would
925 be a natural extension along the lines of [29].

926 We also did not validate the model in true real time. Our held-out 2023 SVB
927 evaluation used contemporaneous news but trained on data up to 2018, which is an honest
928 temporal hold-out but not a real-time deployment. The latency budget for a real-time
929 deployment depends on the use case—pre-trade risk needs sub-second decisions while
930 post-trade compliance can wait minutes—and we do not claim TEGC is optimised for
931 either extreme.
932

933 8. Conclusion

934 This paper introduced TEGC, a framework for learning temporal event-causality
935 graphs from financial news. The framework couples LLM-based event extraction with a
936 neural Hawkes process whose kernels are parameterised by a temporal graph encoder.
937 The result is an interpretable graph whose edges represent learned conditional intensities
938 at which one event type triggers another, parameterised by entities and lag.

939 We made three claims and reported evidence on each. First, **type-level causal-**
940 **temporal associations are learnable from extracted financial events—our learned graph**
941 **recovers relationships consistent with known causal patterns in finance without**
942 **supervision on edges**. Second, the learned structure transfers across episodes—edges
943 trained on news up to 2018 retain meaningful predictive value on held-out cascades from
944 2020 and 2023, with the caveat that the SVB-type fast-cascade regime is harder than the
945 slower-developing episodes. Third, the framework supports interpretable counterfactual
946 cascade simulation, which we illustrated through three case studies.

947 We have tried to be honest about what the framework does and does not do. It learns
948 causal-temporal regularity, not counterfactual causal effects; it inherits extraction errors
949 from its upstream stages; it does not separately identify endogenous market dynamics.
950 We have flagged these limitations explicitly throughout. The framework is most useful as
951 a structured, interpretable scaffold for analyst-driven investigation of contagion
952 mechanisms, not as a black-box predictor.

953 Three natural extensions follow. First, coupling TECG with a structural model of
954 market microstructure would let it separately identify endogenous-mechanism
955 contributions to propagation. Second, **an online variant in the spirit of EvolveGCN [29]**
956 **would address** regime change in real-time deployment. Third, an intervention-data
957 validation—using regulator pre-announcement leaks or natural experiments—would
958 strengthen the causal interpretation of the learned edges. We hope to address each in
959 subsequent work.
960

961 References

- 962 [1] F. Allen and D. Gale, "Financial contagion," *J. Polit. Econ.*, vol. 108, no. 1, pp. 1–33, 2000.
- 963 [2] L. Eisenberg and T. H. Noe, "Systemic risk in financial systems," *Manage. Sci.*, vol. 47, no. 2, pp. 236–249, 2001.
- 964 [3] K. J. Forbes and R. Rigobon, "No contagion, only interdependence: Measuring stock market comovements," *J. Finance*, vol. 57,
965 no. 5, pp. 2223–2261, 2002.
- 966 [4] F. X. Diebold and K. Yilmaz, "Measuring financial asset return and volatility spillovers, with application to global equity
967 markets," *Econ. J.*, vol. 119, no. 534, pp. 158–171, 2009.
- 968 [5] C. A. Sims, "Macroeconomics and reality," *Econometrica*, vol. 48, no. 1, pp. 1–48, 1980.
- 969 [6] H. H. Pesaran and Y. Shin, "Generalized impulse response analysis in linear multivariate models," *Econ. Lett.*, vol. 58, no. 1, pp.
970 17–29, 1998.
- 971 [7] M. Billio, M. Getmansky, A. W. Lo, and L. Pelizzon, "Econometric measures of connectedness and systemic risk in the finance
972 and insurance sectors," *J. Financ. Econ.*, vol. 104, no. 3, pp. 535–559, 2012.
- 973 [8] S. Battiston, M. Puliga, R. Kaushik, P. Tasca, and G. Caldarelli, "DebtRank: Too central to fail? Financial networks, the FED and
974 systemic risk," *Sci. Rep.*, vol. 2, art. 541, 2012.
- 975 [9] F. Feng, X. He, X. Wang, C. Luo, Y. Liu, and T.-S. Chua, "Temporal relational ranking for stock prediction," *ACM Trans. Inf.*
976 *Syst.*, vol. 37, no. 2, art. 27, pp. 1–30, 2019.
- 977 [10] R. Sawhney, S. Agarwal, A. Wadhwa, and R. Shah, "Spatiotemporal hypergraph convolution network for stock movement
978 forecasting," in *Proc. IEEE Int. Conf. Data Mining (ICDM)*, 2020, pp. 482–491.

- [11] A. G. Hawkes, "Spectra of some self-exciting and mutually exciting point processes," *Biometrika*, vol. 58, no. 1, pp. 83–90, 1971.
- [12] R. Trivedi, M. Farajtabar, P. Biswal, and H. Zha, "DyRep: Learning representations over dynamic graphs," in *Proc. Int. Conf. Learning Representations (ICLR)*, 2019.
- [13] E. Rossi, B. Chamberlain, F. Frasca, D. Eynard, F. Monti, and M. Bronstein, "Temporal graph networks for deep learning on dynamic graphs," in *Proc. ICML Workshop on Graph Representation Learning and Beyond*, 2020.
- [14] D. Xu, C. Ruan, E. Korpeoglu, S. Kumar, and K. Achan, "Inductive representation learning on temporal graphs," in *Proc. Int. Conf. Learning Representations (ICLR)*, 2020.
- [15] A. Kulkarni and V. Dogra, "Event-based sentiment analysis of financial news using large language models, retrieval-augmented generation, and graph neural networks," submitted manuscript, 2025.
- [16] A. Kulkarni and V. Dogra, "A comprehensive survey of event extraction methods in natural language processing," in *Proc. Int. Conf. Sustainable Communication Networks and Application (ICSCNA)*, 2024, pp. 925–929.
- [17] F. X. Diebold and K. Yilmaz, "On the network topology of variance decompositions: Measuring the connectedness of financial firms," *J. Econometrics*, vol. 182, no. 1, pp. 119–134, 2014.
- [18] D. Acemoglu, A. Ozdaglar, and A. Tahbaz-Salehi, "Systemic risk and stability in financial networks," *Amer. Econ. Rev.*, vol. 105, no. 2, pp. 564–608, 2015.
- [19] P. Glasserman and H. P. Young, "How likely is contagion in financial networks?," *J. Bank. Finance*, vol. 50, pp. 383–399, 2015.
- [20] X. Ding, Y. Zhang, T. Liu, and J. Duan, "Deep learning for event-driven stock prediction," in *Proc. Int. Joint Conf. Artificial Intelligence (IJCAI)*, 2015, pp. 2327–2333.
- [21] N. Du, H. Dai, R. Trivedi, U. Upadhyay, M. Gomez-Rodriguez, and L. Song, "Recurrent marked temporal point processes: Embedding event history to vector," in *Proc. ACM SIGKDD Int. Conf. Knowledge Discovery and Data Mining*, 2016, pp. 1555–1564.
- [22] H. Mei and J. M. Eisner, "The neural Hawkes process: A neurally self-modulating multivariate point process," in *Adv. Neural Inf. Process. Syst. (NeurIPS)*, vol. 30, 2017, pp. 6754–6764.
- [23] S. Zuo, H. Jiang, Z. Li, T. Zhao, and H. Zha, "Transformer Hawkes process," in *Proc. Int. Conf. Machine Learning (ICML)*, 2020, pp. 11692–11702.
- [24] T. N. Kipf and M. Welling, "Semi-supervised classification with graph convolutional networks," in *Proc. Int. Conf. Learning Representations (ICLR)*, 2017.

- [25] P. Veličković, G. Cucurull, A. Casanova, A. Romero, P. Liò, and Y. Bengio, "Graph attention networks," in Proc. Int. Conf. Learning Representations (ICLR), 2018.
- [26] W. L. Hamilton, R. Ying, and J. Leskovec, "Inductive representation learning on large graphs," in Adv. Neural Inf. Process. Syst. (NeurIPS), vol. 30, 2017, pp. 1024–1034.
- [27] M. Schlichtkrull, T. N. Kipf, P. Bloem, R. van den Berg, I. Titov, and M. Welling, "Modeling relational data with graph convolutional networks," in The Semantic Web, Lecture Notes in Computer Science, vol. 10843, Springer, 2018, pp. 593–607.
- [28] K. Xu, W. Hu, J. Leskovec, and S. Jegelka, "How powerful are graph neural networks?," in Proc. Int. Conf. Learning Representations (ICLR), 2019.
- [29] A. Pareja, G. Domeniconi, J. Chen, T. Ma, T. Suzumura, H. Kanezashi, T. Kaler, T. B. Schardl, and C. E. Leiserson, "EvolveGCN: Evolving graph convolutional networks for dynamic graphs," in Proc. AAAI Conf. Artificial Intelligence, vol. 34, no. 4, 2020, pp. 5363–5370.
- [30] G. Doddington, A. Mitchell, M. Przybocki, L. Ramshaw, S. Strassel, and R. Weischedel, "The Automatic Content Extraction (ACE) program—tasks, data, and evaluation," in Proc. Language Resources and Evaluation Conference (LREC), 2004, pp. 837–840.
- [31] C. Walker, S. Strassel, J. Medero, and K. Maeda, ACE 2005 Multilingual Training Corpus. Philadelphia, PA, USA: Linguistic Data Consortium, 2006.
- [32] H. Ji and R. Grishman, "Refining event extraction through cross-document inference," in Proc. ACL-HLT, 2008, pp. 254–262.
- [33] T. H. Nguyen and R. Grishman, "Event detection and domain adaptation with convolutional neural networks," in Proc. ACL-IJCNLP, 2015, pp. 365–371.
- [34] X. Du and C. Cardie, "Event extraction by answering (almost) natural questions," in Proc. Conf. Empirical Methods in Natural Language Processing (EMNLP), 2020, pp. 671–683.
- [35] J. Devlin, M. W. Chang, K. Lee, and K. Toutanova, "BERT: Pre-training of deep bidirectional transformers for language understanding," in Proc. NAACL-HLT, 2019, pp. 4171–4186.
- [36] Y. Liu, M. Ott, N. Goyal, J. Du, M. Joshi, D. Chen, O. Levy, M. Lewis, L. Zettlemoyer, and V. Stoyanov, "RoBERTa: A robustly optimized BERT pretraining approach," arXiv:1907.11692, 2019.
- [37] A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A. N. Gomez, Ł. Kaiser, and I. Polosukhin, "Attention is all you need," in Adv. Neural Inf. Process. Syst. (NeurIPS), vol. 30, 2017, pp. 5998–6008.

- [38] D. Araci, "FinBERT: Financial sentiment analysis with pre-trained language models," arXiv:1908.10063, 2019.
- [39] T. Loughran and B. McDonald, "When is a liability not a liability? Textual analysis, dictionaries, and 10-Ks," *J. Finance*, vol. 66, no. 1, pp. 35–65, 2011.
- [40] T. B. Brown et al., "Language models are few-shot learners," in *Adv. Neural Inf. Process. Syst. (NeurIPS)*, vol. 33, 2020, pp. 1877–1901.
- [41] H. Touvron et al., "Llama 2: Open foundation and fine-tuned chat models," arXiv:2307.09288, 2023.
- [42] J. Pearl, *Causality: Models, Reasoning, and Inference*, 2nd ed. Cambridge, U.K.: Cambridge Univ. Press, 2009.
- [43] Y. Ogata, "On Lewis' simulation method for point processes," *IEEE Trans. Inf. Theory*, vol. 27, no. 1, pp. 23–31, 1981.